

Notes for ‘Concavity and inflections’

Important Ideas and Useful Facts:

- (i) **Slopes and concavity:** Suppose f is a smooth curve on an interval for which both the derivative f' and the second derivative f'' exist. Call f *concave upwards* or *concave up* if the graph of f lies above all tangent lines (like an ‘upright bowl’), and *concave downwards* or *concave down* if the graph lies below all tangent lines (like a ‘bowl tipped upside-down’). We test for slopes and concavity as follows:
- (a) If $f'(x) > 0$ for all x on an interval then f is sloping upwards on that interval.
 - (b) If $f'(x) < 0$ for all x on an interval then f is sloping downwards on that interval.
 - (c) If $f''(x) > 0$ for all x on an interval then f is concave upwards on that interval.
 - (d) If $f''(x) < 0$ for all x on an interval then f is concave downwards on that interval.
- (ii) **Points of inflection:** A *point of inflection* of a function f is a point on the graph where the concavity changes from concave up to concave down, or from concave down to concave up. At a point of inflection, the tangent line to the curve passes through the point of inflection.

Examples:

1. Consider the curve $y = x^3$. Then $y' = 3x^2$ and $y'' = 6x$, so that both y' and y'' are zero when $x = 0$, and we get the following sign diagrams:

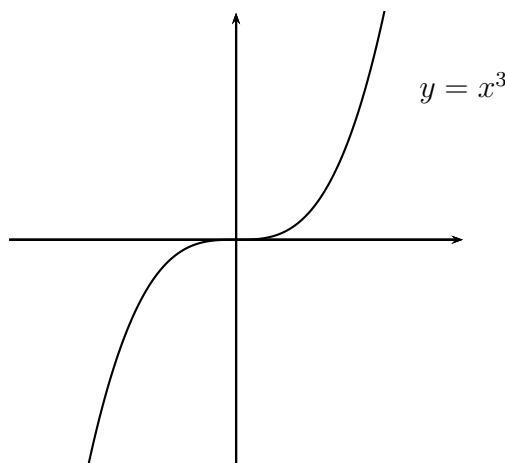
x		0	
y'	+	0	+



x		0	
y''	-	0	+



However, note that y' is positive everywhere other than at $x = 0$, so the function is increasing over all of the real line. In particular, there is no turning point when $x = 0$, even though the derivative becomes zero. There is, however, a point of inflection at $x = 0$, where the curve changes from concave down to concave up.



2. Sketch the curve $y = 9x - x^3$.

Solution: First observe that

$$y = 9x - x^3 = x(9 - x^2) = x(3 + x)(3 - x)$$

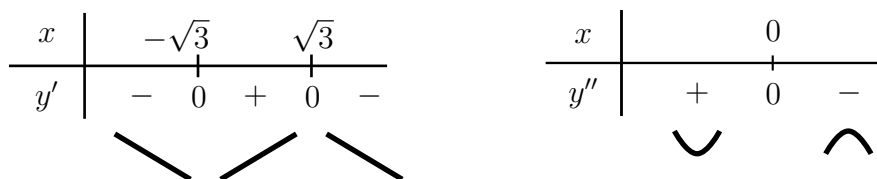
so that $y = 0$ when $x = 0, \pm 3$. We have

$$y' = 9 - 3x^2 = 3(3 - x^2) = 3(\sqrt{3} + x)(\sqrt{3} - x)$$

so that $y' = 0$ when $x = \pm\sqrt{3}$. Further

$$y'' = -6x$$

so $y'' = 0$ when $x = 0$. We get the following sign diagrams for the first and second derivatives:

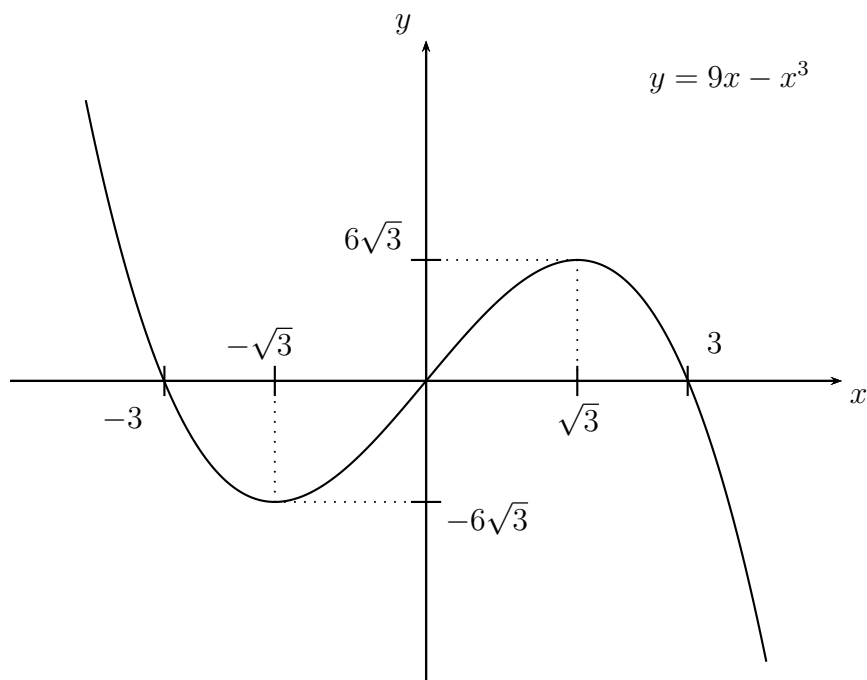


These indicate that there is a local minimum at $x = -\sqrt{3}$, a local maximum at $x = \sqrt{3}$ and a point of inflection when $x = 0$, at which point the curve changes from concave up to concave down. The original factorisation of y tells us also that the curve crosses the x -axis at $x = -3, 0$ and 3 . The local maximum is

$$y(\sqrt{3}) = 9\sqrt{3} - 3\sqrt{3} = 6\sqrt{3},$$

and the local minimum is

$$y(-\sqrt{3}) = -9\sqrt{3} + 3\sqrt{3} = -6\sqrt{3}.$$



3. Sketch the curve $y = x^{2/3}$.

Solution: First observe, using the rule for differentiating a power of x (by bringing the exponent to the front and creating a new exponent by subtracting one), that

$$y' = \frac{2}{3}x^{-1/3} = \frac{2}{3x^{1/3}}.$$

Differentiating again, we get

$$y'' = \left(\frac{2}{3}\right)\left(-\frac{1}{3}\right)x^{-4/3} = -\frac{2}{9x^{4/3}} = \frac{-2}{9(x^{2/3})^2},$$

written this way just to emphasise that the denominator is always positive for $x \neq 0$. We get the following sign diagrams, noting that both y' and y'' are undefined for $x = 0$:



Note that $y = 0$ when $x = 0$, and the curve, which includes the origin, is continuous, that is, can be drawn without lifting the pen off the paper. The sign diagram for the derivative tells us that there must be a global minimum at $x = 0$. The sign diagram for the second derivative tells us also that the curve is concave down on both the negative and positive halves of the real line.

To understand the slope near the origin, we can observe the following one-sided limits:

$$\lim_{x \rightarrow 0^+} y' = \lim_{x \rightarrow 0^+} \frac{2}{3x^{1/3}} = \infty \quad \text{and} \quad \lim_{x \rightarrow 0^-} y' = \lim_{x \rightarrow 0^-} \frac{2}{3x^{1/3}} = -\infty,$$

noting that as x gets close to zero from the positive side, then y' gets arbitrarily large and positive, whilst if x get close to zero from the negative side, then y' gets arbitrarily large and negative. The first limit tells us that the tangent lines to the curve, as we move from the right towards zero, become closer and closer to being vertical, using steep positive slopes. The second limit tells us that, as we move from the left towards zero, the tangent lines again become closer and closer to being vertical, but using steep negative slopes. The curve, sketched below, has a *cusp* at the origin.

